What does Plate Glass Reveal about Camera Calibration?

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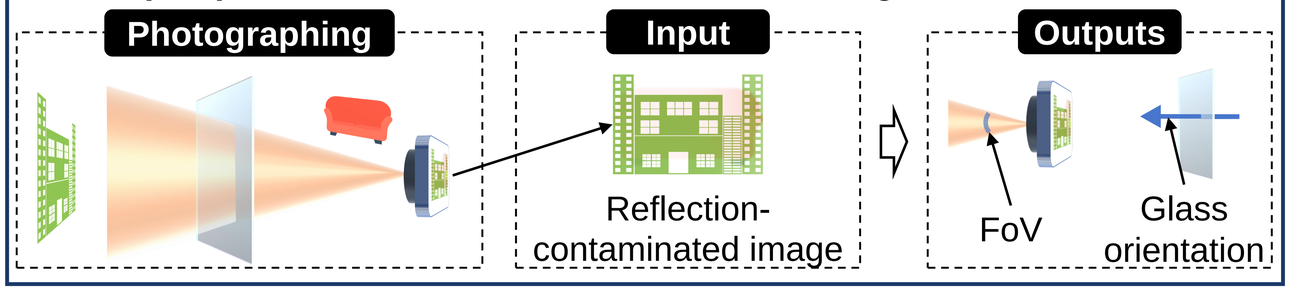
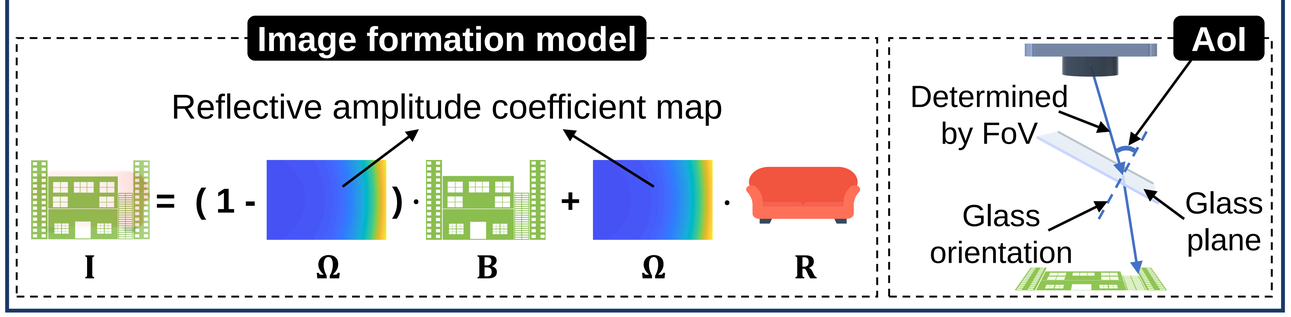
PROBLEM DESCRIPTION

- Take a **reflection-contaminated image** as the input, calibrate the **field of view (FoV)** of the camera and the **orientation of the glass**.



OBSERVATIONS

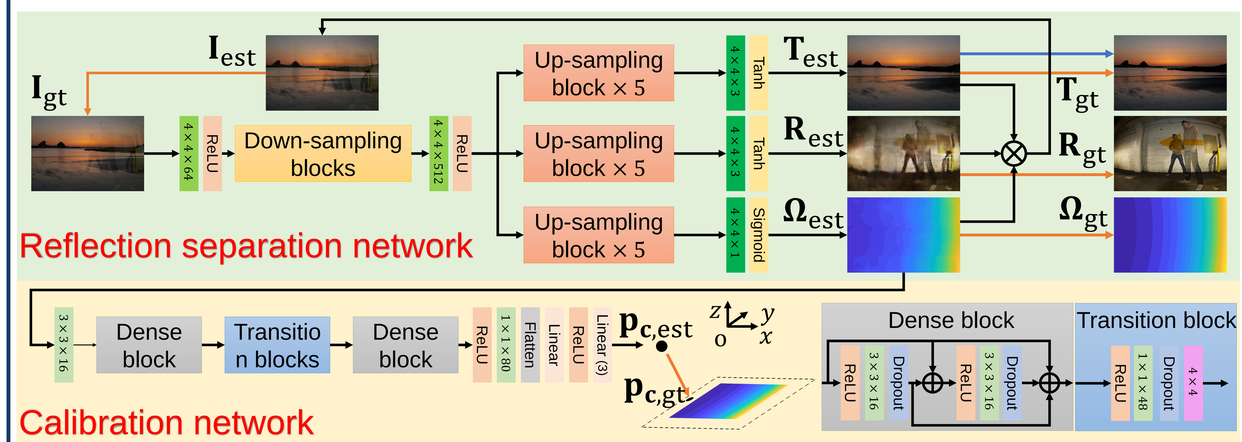
- The **reflective amplitude coefficient** is monotonically increasing regarding to the **angle of incidence** (**AoI**, regarding to the glass plane).
- Each **AoI** is determined by the **FoV** and the **glass orientation**.



CONTRIBUTIONS

- Present the first work to calibrate camera parameters in the context of glass reflection.
- Propose a learning-based method to jointly estimate the reflective amplitude coefficient map and camera parameters.
- Show the proposed method is free from image contents, benefits to solving problems of reflection removal and panorama estimation from a single image.

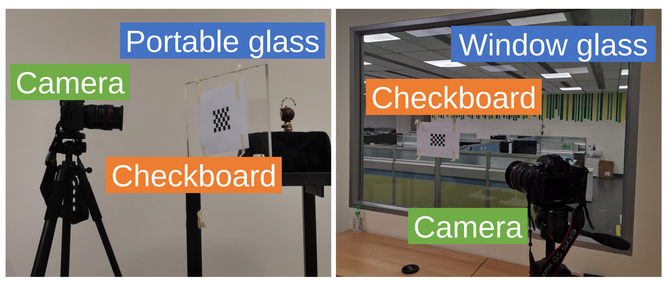
TWO-STEPS SOLUTION



$$\text{FoV} = 2\arctan\left(\frac{w}{2\mathbf{p}_{\mathbf{c,est}}(3)}\right), \qquad \text{Glass orientation} = \frac{\mathbf{p}_{\mathbf{c,est}}}{\|\mathbf{p}_{\mathbf{c,est}}\|_2}$$

EXPERIMENTS

Validation data collection



Controlled Wild

Training data synthesizing

$$\mathbf{I} = (1 - \mathbf{\Omega}) \cdot \mathbf{B} + \mathbf{\Omega} \cdot \mathbf{R}$$

$$\mathbf{\Omega} = \mathrm{F}(\theta, \kappa) = \frac{1}{2}\left(\frac{2\mathcal{R}_{\perp}^{s}}{1+\mathcal{R}_{\perp}^{s}} + \frac{2\mathcal{R}_{\parallel}^{s}}{1+\mathcal{R}_{\parallel}^{s}}\right)$$

Camera calibration

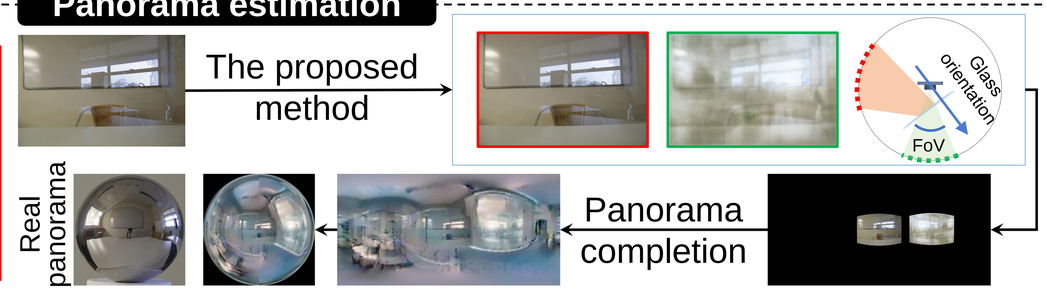
Free from Image Content

FoV estimation	Sample 1	Sample 2
GT	58.7°	19.5°
Image content based method	22.9°	16.4°
Ours	**63.5°**	39.4°
Fusion	31.0°	**20.0°**

Reflection removal (SSIM)

Better performance

Dataset	ZN18	YM19	WS19	WY19	WT19	Ours
SIR²	0.8453	0.8547	0.9003	0.8885	0.8550	0.8895
ZN18-DATA	0.8210	0.7220	0.7654	0.7193	0.8180	0.7510
PB19-DATA	0.7970	0.8308	0.8580	0.8713	0.8359	0.8736
CONTROL	0.8115	0.7877	0.8283	0.7950	0.8004	0.8445

Panorama estimation

Alleviate the ill-posedness